

CPSC 3720 – Assignment 4: Real-Time Software Engineering & Project Planning

Due Date : Dec 2nd , 2025 on Moodle

Total Gades : 100

Objective

This assignment builds on Chapter 21 (Real-Time Systems), Chapter 23 (Project Planning), and Chapter 24 (Cost Estimation) of the Sommerville textbook, Lecture material covered in the week of Nov 7th to Nov 21st , 2025. It is designed to integrate with your ongoing project. In case it is not synchronizing with your project you can take any example project to discuss these. You will apply real-time software engineering principles and project planning techniques to enhance your system's responsiveness, reliability, and manageability.

Note

If the topic of this assignment correlates with the work you submitted in Assignment 2, Assignment 3, or your final project, you are encouraged to build upon that work for this assignment else work on any related project.

Part A: State-Based Modeling (15 Marks)

1. Identify a subsystem in your project that can be modeled using a state-based approach (e.g., user interaction, sensor control, message handling).
2. Create a state diagram showing all relevant states and transitions.
3. Describe how this model improves system clarity and reliability.

Part B: Real-Time Constraints (15 Marks)

4. Identify any real-time constraints in your project (e.g., timing, latency, periodic data collection).
5. Explain how you plan to schedule periodic processes to meet these constraints.
6. Justify your scheduling decisions using examples from Chapter 21.

Part C: Environmental Control Pattern (15 Marks)

7. Apply the Environmental Control pattern to a relevant part of your system (e.g., temperature, light, or resource regulation).
8. Describe the sensors, controllers, and actuators involved.
9. Provide a control loop or logic diagram.

Part D: Stimuli and Responses (15 Marks)

10. Identify stimuli your system must respond to (e.g., user input, sensor data, external signals).
11. Describe the responses and how they are handled in real-time.
12. Discuss how your design ensures safety and correctness.

Part E: Project Planning (25 Marks)

13. Develop a Work Breakdown Structure (WBS) for your project.
14. Create a Gantt chart or timeline showing major tasks and milestones.
15. Identify risks and propose mitigation strategies.
16. Estimate effort and resources using techniques from Chapter 23.
17. Discuss how planning impacts your ability to meet real-time constraints.

Part F: Reflection (15 Marks)

18. Write a short reflection (200–300 words) on how integrating real-time software engineering principles and project planning has influenced your project design and development.

Grading Rubric

Criteria	Excellent (A)	Good (B)	Fair (C)	Poor (D/F)
State-Based Modeling	Clear, complete diagram; insightful explanation	Mostly clear diagram; minor gaps	Diagram present but lacks clarity	Missing or incorrect
Real-Time Constraints	Thorough identification and justified scheduling	Adequate identification; some justification	Limited identification; weak justification	Missing
Environmental Control Pattern	Well-applied pattern with detailed diagrams	Applied pattern with minor gaps	Basic application; lacks detail	Missing
Stimuli and Responses	Comprehensive and safety-focused	Adequate coverage; minor gaps	Limited coverage; unclear safety	Missing
Project Planning	Detailed WBS, Gantt chart, risk analysis, resource estimates	Mostly complete; minor gaps	Partial planning elements	Missing
Reflection	Deep insights; well-written	Good insights; minor issues	Basic reflection; lacks depth	Missing

Grade Distribution

- Part A: 15
- Part B: 15
- Part C: 15
- Part D: 15
- Part E: 25

- Part F: 15